

## ENCN423 / 627 Formula Sheet

$$V_{MPP-min} = V_{MPP} \times \left[ 1 + \left( (T_{max} + T_{add} - T_{STC}) \times \frac{T_k}{100} \right) \right]$$

$$V_{OC-max} = V_{OC} \times \left[ 1 + \left( (T_{min} - T_{STC}) \times \frac{T_k}{100} \right) \right]$$

$$E_{\text{useful}} = C \cdot V_{\text{nominal}} \cdot (\text{SoC}_{\text{max}} - \text{SoC}_{\text{min}})$$

$$V_{\text{nominal}} = \frac{V_{\text{max}} + V_{\text{min}}}{2}$$

$$acceleration = \frac{v_2 - v_1}{t}$$

$$distance = \frac{v_1 + v_2}{2} \cdot t$$

$$\Delta KE = \frac{1}{2} m (v_2^2 - v_1^2)$$

$$W_{\text{roll}} = C_{\text{roll}} \cdot m \cdot g \cdot d$$

$$W_{\text{drag}} = \frac{1}{8} \cdot \rho \cdot C_{\text{drag}} \cdot A \cdot \frac{v_2^4 - v_1^4}{a}$$

$$LCOF = \frac{CAPEX \cdot crf + OPEX + \sum (LCOI_n \cdot Input_n)}{Output_{\text{fuel}}}$$

$LCOI_n$  – Levelized Cost of Input  $n$  (e.g.,  $LCOE$ ,  $LCOH_2$ , cost of  $\text{CO}_2$ , heat, etc.).

$$crf = \frac{r \cdot (1 + r)^{\text{Lifetime}}}{(1 + r)^{\text{Lifetime}} - 1}$$

$$C = \frac{E_{\text{total}}}{V_{\text{nominal}}}$$